

Infinity Hotel Check-In Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	infinity-hotel-check-in

Big idea: Mathematical concepts can be coherent even when they violate ordinary experience.

One-Minute Teacher Brief

Students act out Hilbert's Hotel to examine why mathematical infinity resists everyday intuition.

Student Prompt

Inspect Infinity Hotel Check-In Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Infinity Hotel Check-In Stimulus A	Student-facing stimulus 1: One new guest arrives at a full infinite hotel; every current guest moves by $n \rightarrow n + 1$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Infinity Hotel Check-In Stimulus B	Student-facing stimulus 2: Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Infinity Hotel Check-In Reveal / Twist	Student-facing stimulus 3: Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical concepts can be coherent even when they violate ordinary experience.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Infinity Hotel Check-In Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Infinity Hotel Check-In Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Can mathematics know beyond intuition?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how mathematical concepts can be coherent even when they violate ordinary experience.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Can mathematics know beyond intuition?

Setup Checklist

- Use numbered cards on desks or around the room to represent hotel rooms.
- Make clear when students are reasoning about a formal model rather than a physically possible hotel.

Example Stimuli or Materials

- One new guest arrives at a full infinite hotel; every current guest moves by $n \rightarrow n + 1$.
- Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified.

Resources To Use

- Resource pack: <resources/infinity-hotel-check-in/index.html>
- Card deck CSV: <resources/infinity-hotel-check-in/cards.csv>
- Visual prompt: <resources/infinity-hotel-check-in/visual-prompt.svg>
- Room card set and move-rule card.
- Reflection sheet: what felt impossible, what the rule showed, and what the model assumes.

Run Sequence

1. Assign students numbered hotel rooms and announce that every room is occupied.
2. Introduce one new guest and ask whether the hotel can make room.
3. Use the rule $n \rightarrow n + 1$ so each guest moves to the next room.
4. Discuss why the model works for countably infinite rooms but not for a finite hotel.
5. Debrief the difference between everyday imagination and formal definition.

Debrief Moves

- Knowledge question: Can mathematics produce knowledge about things we cannot imagine concretely?
- Why can a coherent formal system feel impossible?

- What is the role of intuition in mathematical knowledge?

Teacher Quick Script

- Do not ask whether a real hotel could do this; ask whether the formal rule is coherent.
- Where did your everyday intuition stop helping?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...